CORRECTIVE AND PREVENTIVE ACTION (CAPA) REPORT

CAPA Number	**QA-CAPA-2024-0118**	**Product**	**VascuFlow CVC-14G**
Date Opened	2024-04-02	Date Closed	2024-10-15
CAPA Owner	Quality Engineering	Status	Closed - Effective
Site	Manufacturing Site 3	Risk Class	Class II

1.0 BACKGROUND

The VascuFlow Central Venous Catheter (CVC-14G) is a single-use, percutaneously inserted catheter indicated for short- to medium-term central venous access. The device has been commercially distributed since 2019 across North America, EMEA, and APAC. This report documents the investigation, root cause analysis, and corrective actions taken in response to an increase in insertion-related complaints observed beginning in Q1 2024.

2.0 COMPLAINT SUMMARY

During Q1, the Quality organization received 14 product complaints related to the VascuFlow Central Venous Catheter (Model CVC-14G), representing a complaint rate of 0.08% of units distributed. Eleven complaints concerned difficulty with guidewire advancement during insertion. Three complaints reported minor leakage at the luer connector under high infusion pressure. No complaints met the criteria for a reportable adverse event under 21 CFR Part 803.

Table 1: Complaints by Lot

Lot Number	Complaints	Units Shipped	Rate (%)	Status
CVC-2024-0298	1	5100	0.02	Investigated - Unrelated
CVC-2024-0305	2	4950	0.04	Investigated - Unrelated
CVC-2024-0311	5	4800	0.10	Investigated - Root Cause
CVC-2024-0322	3	5050	0.06	Investigated - Root Cause
CVC-2024-0341	4	4700	0.09	Investigated - Root Cause
CVC-2024-0356	3	4900	0.06	Investigated - Root Cause
CVC-2024-0389	6	5200	0.12	Investigated - Root Cause
CVC-2024-0402	1	5000	0.02	Investigated - Unrelated

3.0 ROOT CAUSE ANALYSIS AND CORRECTIVE ACTION

Investigation traced the guidewire advancement issue to a supplier change in the catheter's inner lumen coating, introduced in lot range CVC-2024-0311 through CVC-2024-0389. The coating's coefficient of friction exceeded the design specification of 0.20. A fishbone analysis identified the coating supplier's curing process temperature as the primary contributing factor; the new supplier's oven profile ran approximately 8 degrees C cooler than the qualified process, resulting in incomplete cross-linking of the coating polymer.

A designed experiment (DOE) varying oven temperature, dwell time, and coating solution viscosity was conducted across three pilot runs to confirm the causal relationship. The DOE confirmed that oven temperature was the dominant factor, accounting for approximately 78% of the variation in measured coefficient of friction, with dwell time contributing a secondary, smaller effect. Coating solution viscosity was not found to be statistically significant within the tested range. Statistical process control charts for the affected lot range showed the friction coefficient trending upward beginning with the first lot manufactured after the supplier's oven was recommissioned following a scheduled maintenance event, further corroborating the oven temperature hypothesis.

Root Cause Verification: A confirmation run was performed at the qualified oven temperature setpoint using the new supplier's coating solution; the resulting coefficient of friction returned to within specification (mean 0.14, all samples below the 0.20 limit), confirming that oven temperature, not the coating material itself, was the true root cause.

Corrective Actions:

1. Reverted to the qualified coating supplier's oven temperature setpoint for all lots produced after April 2024.
2. Added an in-process friction test to the lumen coating step, with a hard stop at a coefficient of 0.20.
3. Initiated a voluntary field correction notice for affected lots still in distributor inventory.
4. Updated the incoming supplier qualification protocol to include oven profile verification as a standard acceptance test.
5. Added a quarterly oven temperature calibration check to the preventive maintenance schedule for all coating equipment.
6. Retrained coating line operators on the revised process control limits and the escalation procedure for out-of-trend friction readings.

4.0 RISK ASSESSMENT

A risk assessment was performed per ISO 14971 to evaluate the clinical impact of the guidewire advancement difficulty. Increased insertion time and the potential for a failed first-pass attempt were identified as the primary hazards. Severity was assessed as Minor (no reported patient harm across all complaints), and probability of occurrence was reduced from Occasional to Remote following implementation of the corrective actions. The overall risk was determined to remain within acceptable limits per the product's existing risk management file, and no design change to the risk controls was required beyond the manufacturing process correction described in Section 3.0.

5.0 VERIFICATION OF EFFECTIVENESS

An effectiveness check was performed six months after implementation of the corrective actions, covering lots produced from May 2024 through October 2024. In-process friction testing showed 100% compliance with the 0.20 specification limit across all 26 lots tested in the period, with a mean coefficient of friction of 0.145 (versus 0.21 during the affected period). Zero guidewire advancement complaints were received for product manufactured after the corrective action implementation date. The CAPA effectiveness check was determined to be PASS, and the CAPA was formally closed on 2024-10-15.

6.0 REGULATORY REPORTING ASSESSMENT

Per 21 CFR Part 803 and the site's Medical Device Reporting procedure, none of the 25 complaints associated with this CAPA met the criteria for a reportable event, as none resulted in death, serious injury, or malfunction that would be likely to cause or contribute to death or serious injury if it were to recur. This assessment was reviewed and concurred with by Regulatory Affairs.

Approval

Role	Name	Signature	Date
Quality Engineer (Prepared By)			
Quality Manager (Reviewed By)			
RA Director (Approved By)			